Quantum Machine Learning for Early Disease Prediction

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Abstract—Early detection of diseases is an extremely important area of research in artificial intelligence in healthcare. By applying principles of quantum computing and machine learning, Quantum Machine Learning (QML) has become a promising research area in order to develop models that can accurately predict diseases. The paper presents a review of recent literature in the field of QML for disease prediction. The literature review covers various QML methodologies, datasets, results and limitations reported. It has been shown in previous work that disease predictions can be performed through various QML methods including Quantum Support Vector Machines (QSVM), Quantum Neural Networks (QNN), Variational Quantum Circuits (VQC), quantum ensembling and optimizations. An experimental framework was developed implementing QML techniques on the UCI Heart Disease dataset. Data pre-processing, scaling, PCA and quantum feature maps were implemented, alongside a fidelity quantum kernel and Quantum Support Vector Classifier (QSVC). A classical Support Vector Classifier (SVC) was used to compare performance. Limitations of the methodology were also noted including small dataset size, reliance on quantum simulators, feature reduction and limited quantum resources. This paper provides a structured foundation for further research into developing scalable, reliable and applicable QML systems in healthcare.

Index Terms—Quantum Machine Learning, Quantum Computing, Early Disease Detection, Heart Disease, Healthcare, QSVM, QSVC, Fidelity Quantum Kernel, PCA, Clinical Decision Support.

I. INTRODUCTION

The early prediction of diseases is becoming increasingly important. With the ability to identify risk factors associated with diseases early on, the patient can receive clinical attention at a much earlier stage. Modern healthcare systems can provide a wealth of information such as electronic health records, medical imaging, wearable technology and genomic resources. However, often this data can be challenging to process due to being heterogenous, high dimensional. Machine learning techniques have become a valuable tool for diagnosis, prediction of risk, medical imaging and decision making. Traditional models still pose challenges once healthcare data becomes complex.

Quantum Computing provides an alternative approach to computing. This can be combined with Machine Learning techniques to create Quantum Machine Learning. This has been studied in problems like classification, feature representation, optimisation and prediction. Recently, there have been some papers looking at using different models (for example QSVM, QNN and VQC) to perform tasks related to disease in Healthcare. However, many of these experiments are performed on simulators and on small datasets. This makes it difficult to know if we can expect any improvement when implementing these models in real world scenarios.

In this paper we review recent advances in Quantum Machine Learning (QML) for early disease detection and identify common obstacles and gaps. An experimental QML framework is created using the UCI Heart Disease dataset to assess Quantum Kernel based disease prediction. This uses data preprocessing, feature scaling, Principal Component Analysis (PCA), quantum feature encoding, Fidelity Quantum Kernel and Quantum Support Vector Classification (QSVC). A classical Support Vector Classifier (SVC) was used for comparison. While no quantum advantage was shown, this study has assessed how QML can be applied in practice using a controlled experiment. The limitations and need for further validation were also discussed.

II. LITERATURE REVIEW

A. QML in Healthcare

QML is a new area of research at the intersection of quantum computing with machine learning, applied to healthcare analytics. This can be seen through review papers discussing its uses within medical imaging, disease prediction, individualised treatment, genomics and aiding decisions. However, current quantum hardware has many limitations including noise, low qubit numbers and imperfect gates making it difficult to implement large-scale healthcare experiments. Reviews on digital health also highlight a lack of real-world evidence and consistent results when comparing against strong classical baselines [1][3][8].

B. QML for Disease Prediction

One of the key applications where quantum machine learning can make an impact, is through disease prediction using classification models. In terms of cardiovascular research, quantum support vector methods, Pegasos QSVC and quantum neural networks have been explored [2] [5] [6]. A QNN model achieved an accuracy of 97.31% for a multi-class ECG image task [5]. The quantum classification models also showed an increase in performance when comparing to certain classical configurations. [5]. This gives us promising results however we must consider the conditions of the experiments, nature of the dataset and the fact they were run on simulators. Overall the literature provides a good basis for exploring quantum kernel based methods for predicting heart disease using structured clinical features.

C. Current Challenges

Four issues were evident throughout the literature review. Firstly, many studies used quantum simulators rather than quantum processors. Secondly, there was often limited data with specific disease information. Thirdly, encoding and reducing data can become an issue when converting classical healthcare data to a format that can be used by quantum circuits. Finally, integrating quantum computing into hospitals has many considerations including interoperability, patient confidentiality, compliance, interpretability and clinical validation [7] [8].

D. Research Trends

The current research trends focus on hybrid quantum-classical algorithms. This includes using a classical algorithm for data preparation and optimisation, but using quantum circuits for select parts of the computation. Other trends include improvements to feature maps, variational circuits, quantum kernels, noise reduction and comparison to classical algorithms. However, there is increasing focus on using larger datasets and real-world hardware. Consistent proof of a practical quantum advantage in the healthcare setting has yet to be demonstrated [7][8].

III. COMPARATIVE ANALYSIS OF REVIEWED STUDIES

TABLE I. COMPARISON OF RECENT QML STUDIES FOR EARLY DISEASE PREDICTION

Ref.	Disease / Application	QML Algorithm	Key Finding	Limitation	Future Work
[1]	Healthcare	Multiple QML approaches	Reviews emerging QML applications across healthcare and identifies potential areas for future research	Limited real-world clinical validation and hardware maturity	Evaluate QML on larger datasets and real quantum hardware
[2]	Heart disease	QML + QPSO	QML with preprocessing and QPSO achieved 96.7% accuracy, outperforming the evaluated classical MLP model	Dataset and computational setting limit generalization	Evaluate with broader datasets and realistic quantum environments
[3]	Medical decision support	QC + ML / QML approaches	Reviews how quantum computing and ML can support diagnosis, prediction, optimization, and clinical decision-making	Many applications remain experimental or conceptual	Develop real-world case studies and clinically validated implementations
[4]	Disease detection and prediction	Various QML approaches	Surveys QML applications for disease detection and prediction across multiple healthcare domains	Limited hardware maturity and practical validation	Develop benchmarks, improved feature encoding, and hardware-based evaluation
[5]	Cardiovascular disease	QSVC, Pegasos QSVC, QNN	QNN achieved 97.31% accuracy and outperformed the classical CNN; QSVC also outperformed SVC.	Evaluation depends on the selected dataset and experimental setup	Test on broader cardiovascular datasets and quantum hardware
[6]	Heart disease	Quantum Ensemble Learning (QuEML)	QuEML achieved a 0.6% accuracy improvement and reduced average training time compared with the evaluated classical approaches	Improvement was modest and evaluation used a Kaggle dataset	Validate on larger clinical datasets and optimize the ensemble
[7]	QML theory and applications	QNN, QBoost and related methods	Identifies important QML challenges including trainability, noise, scalability, and achieving quantum advantage	Noise, scalability, and hardware limitations remain unresolved	Develop error mitigation, fault-tolerant methods, and improved algorithms
[8]	Digital health	Various QML models	Systematic review finds growing QML research in digital health but limited consistent evidence of practical quantum advantage	Few realistic hardware evaluations and inconsistent benchmarking	Improve hardware-based evaluation, robustness, scalability, and standardized benchmarking

IV. RESEARCH GAP

The literature suggests that QML can be promising for certain disease prediction tasks, however it is not currently at the stage where it can be practically deployed into healthcare. A key issue is that most studies assess a QML algorithm in isolation without considering the full pipeline. Whilst many studies will look to apply a classifier to a specific disease type or dataset, in practice a healthcare solution would require consideration of the full pipeline including data pre-processing, feature representation, prediction and interpretation.

Finally, issues related to validating and benchmarking the models exist. While using simulators is useful for developing and testing quantum models, it does not account for all aspects that may be present when running on real hardware. This could include noise, calibration drift and connectivity. Furthermore, the small/ special data sets may not provide information regarding the generalisability of the findings to other patient populations. Inconsistencies have been observed in standardised methodology used to evaluate the QML models compared to classical baselines.

In order to close these gaps, this study explores the use of a quantum kernel based methodology to predict heart disease using the UCI Heart Disease dataset. The experiment consists of a series of classical steps (preprocessing and PCA for dimensionality reduction). These steps are then followed by a quantum feature map where PCA features are mapped using the Fidelity Quantum Kernel. Finally, QSVC is used to classify the data. An SVC model was also trained on the same training subset using classical methods for comparison. This serves to provide an insight into the suitability of QML for heart disease prediction.

V. PROPOSED FRAMEWORK

This proposed system uses a combination of classical preprocessing and quantum machine learning to predict heart disease. First, the data will be gathered from the UCI Heart Disease dataset. The data will then be preprocessed by handling any missing data. This will also convert the clinical variables into a numerical format. Finally, they will be standardised. PCA will then be applied to the data to reduce the original number of features to 4 so that they can be used in a quantum system.

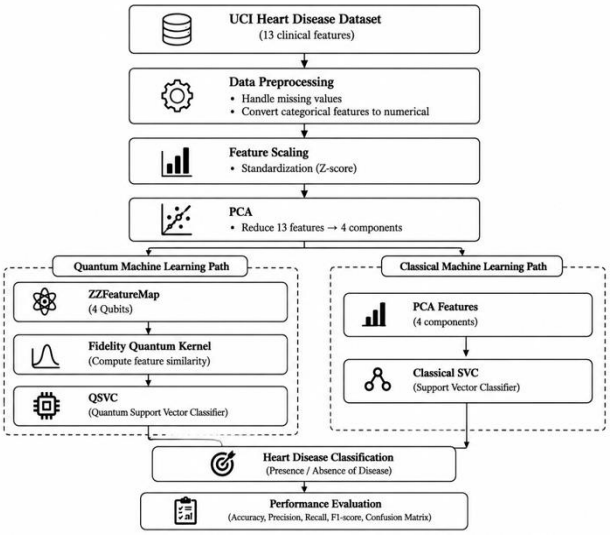
The reduced data can be mapped to the quantum space using a quantum feature map, this example uses the ZZFeatureMap. The Fidelity Quantum Kernel can then be used to determine the similarities between each of the data points in the quantum feature space. This kernel can be given to a Quantum Support Vector Classifier (QSVC) to classify whether a patient has heart disease. The classical Support Vector Classifier (SVC) is also tested using the same matched training subset and test set.

The prediction stage produces a classification indicating the presence or absence of heart disease. The resulting performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix. The system is intended as an experimental decision-support framework rather than an autonomous clinical diagnosis system.

Fig. 1. Proposed conceptual framework for quantum machine learning-based early disease prediction.

VI. IMPLEMENTATION AND EXPERIMENTAL RESULTS

The proposed framework was implemented as a proof-of-



concept binary disease-classification pipeline using the **UCI Heart Disease dataset**. The dataset contains **303 samples with 13 clinical features** and a binary target indicating the presence or absence of heart disease. The implementation used Python, scikit-learn for preprocessing and the classical baseline, and Qiskit Machine Learning for the quantum-kernel model. The workflow used an **80:20 stratified train/test split**, resulting in **242 training samples and 61 test samples**. The clinical features were standardized and Principal Component Analysis (PCA) was applied to reduce the 13-dimensional feature space to four components for quantum processing.

Fig. 2 shows the implemented system architecture, including the classical SVC baseline and the quantum-kernel classification stage.

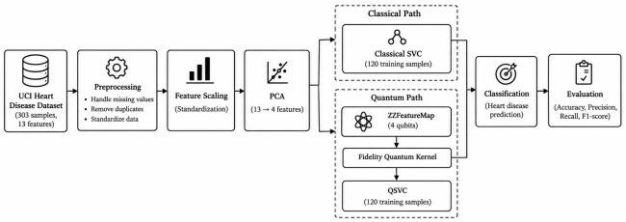


Fig. 2. Implemented system architecture for the QML-based heart disease prediction pipeline.

The implementation procedure is displayed in Fig. 3. 120 training examples were chosen from the PCA samples using stratified sampling. Both the classical SVC and quantum SVC models used this training data. The same test set of 61 samples was also used to evaluate the performance. The quantum model used a 4 qubit ZZFeatureMap with a Fidelity Quantum Kernel. The matching training data was used due to the high computational cost of simulating a fidelity quantum kernel.

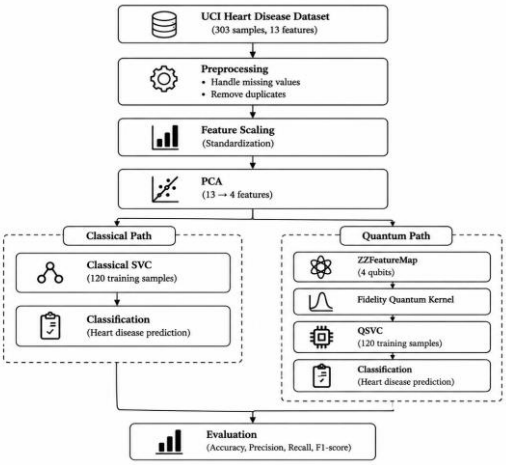


Fig. 3. Flowchart of the implemented experimental workflow.

The results in Table II show that the classical SVC achieved higher performance across all evaluation metrics compared with QSVC. The QSVC obtained an accuracy of 44.26%, while the classical SVC achieved 88.52%. The lower QSVC performance can be attributed to the limited training subset, PCA-based feature reduction, quantum feature-map configuration, and the computational constraints of quantum-kernel simulation. Therefore, the results demonstrate the feasibility of implementing a quantum-kernel-based approach rather than claiming a quantum advantage.

TABLE II. PERFORMANCE COMPARISON OF CLASSICAL SVC AND QSVC

Metric	Classical SVC	QSVC
Accuracy	**88.52%**	**44.26%**
Precision	**86.21%**	**31.25%**
Recall	**89.29%**	**17.86%**
F1-score	**87.72%**	**22.73%**

The QSVC confusion matrix in Fig. 4 presents the classification results obtained on the 61-sample test set. The matrix provides the numbers of correctly and incorrectly classified instances for the two heart disease classes. The corresponding accuracy, precision, recall, and F1-score are reported in Table II. The results are interpreted within the controlled experimental setting, considering the use of PCA-based feature reduction, a limited matched training subset, and quantum-kernel simulation.

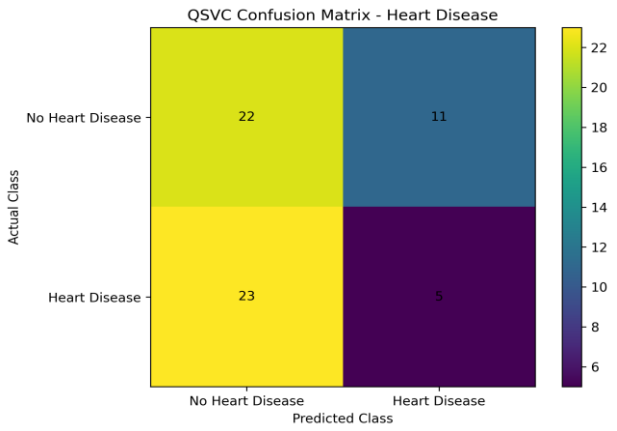


Fig. 4. Confusion matrix of the QSVC model on the test dataset.

Fig. 5 compares the four evaluation metrics for the classical SVC and QSVC models. The experiment demonstrates an end-to-end QML workflow for heart disease classification and provides a controlled comparison between classical and quantum-kernel approaches using the same training and test samples. The results also highlight the practical effects of feature reduction, limited training data, and quantum-kernel simulation on model performance.

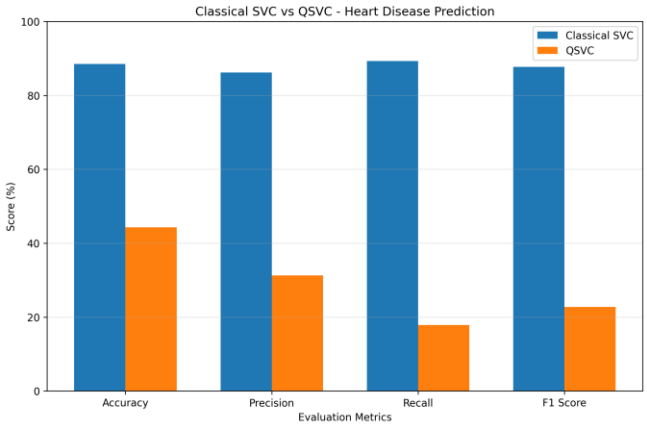


Fig. 5. Performance comparison of Classical SVC and QSVC.

VII. DISCUSSION

The experimental results demonstrate that QML can be applied to structured heart disease data using a quantum-kernel-based classification approach. However, the results also show that QML performance should not be assumed to be superior to classical machine learning. The observed performance depends on factors such as dataset size, feature reduction, quantum feature-map design, kernel configuration, training-set size, and the use of quantum simulation. Therefore, the results should be interpreted as an experimental evaluation of QML rather than evidence of universal quantum superiority.

We can implement a full pipeline including preprocessing, scaling features, PCA dimensionality reduction, quantum feature mapping, Fidelity Quantum Kernels and QSVC classification. The SVC and QSVC algorithms use the same training subset and test set, which allows us to compare the two approaches more accurately. However, due to the small size of the dataset, and the cost of simulating quantum kernels we cannot run a large scale experiment.

This suggests future work in healthcare on QML should consider performance metrics such as data quality, features, computational demands, hardware limitations, explainability, privacy, and clinical testing. This provides a realistic basis for further research in the field of quantum machine learning for early disease prediction.

VIII. LIMITATIONS AND FUTURE WORK

The implemented system is a proof-of-concept experimental model and has not been presented as a clinically validated diagnostic system. The evaluation is based on the UCI Heart Disease dataset, which contains a limited number of samples and clinical features. The use of PCA to reduce the feature space to four components may also affect the information available to the classifier. In addition, the quantum-kernel model is evaluated using simulation, which introduces computational limitations and does not represent all constraints of physical quantum hardware.

Further research is needed to explore this approach using larger and more varied clinical data, different quantum feature maps and kernels for comparing. The experiments can be expanded from ideal to noisy simulations and where possible physical quantum processors. Additional research could look into efficient feature encodings, noise reduction techniques, interpretability, hybrid quantum-classical models and integration with clinical information systems.

IX. CONCLUSION

Quantum Machine Learning is an emerging field which can help understand disease prediction from structured medical data. Research has shown that some Quantum Machine Learning models can have promising results. This includes Quantum Support Vector Machines and Quantum Neural Networks. However, there are still issues with hardware limitations, size of data and validation of results. In this study a pipeline was created to predict heart disease using a Quantum Machine Learning model. The UCI Heart Disease dataset was used along with classical data processing and dimensionality reduction using Principal Component Analysis. A Quantum Support Vector Machine was also used with a ZZFeatureMap, Fidelity Quantum Kernel and QSVC. A classical Support Vector Machine was also used using the same training subset and testing set.

This work demonstrates how a quantum kernel method can be applied and tested in practice for predicting heart disease. The experimental results do not demonstrate a general advantage for quantum computation. To make progress towards practical QML applications in medicine, we need large varied data sets, clear benchmarks, real-world hardware testing, efficient quantum encodings, and strong collaboration between quantum scientists, medical experts and biomedical specialists.

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